ELS2 - Mé422

CALCUL NUMERIQUE EN MECANIQUE DES STRUCTURES

TP 1 : CALCUL LINEAIRE DE POUTRE



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1 Model Justification

Consider a rectangular hollow beam of length L = 2m, thickness 4mm, fixed along its entire cross-section at x=0, whose cross-section is defined as follows :

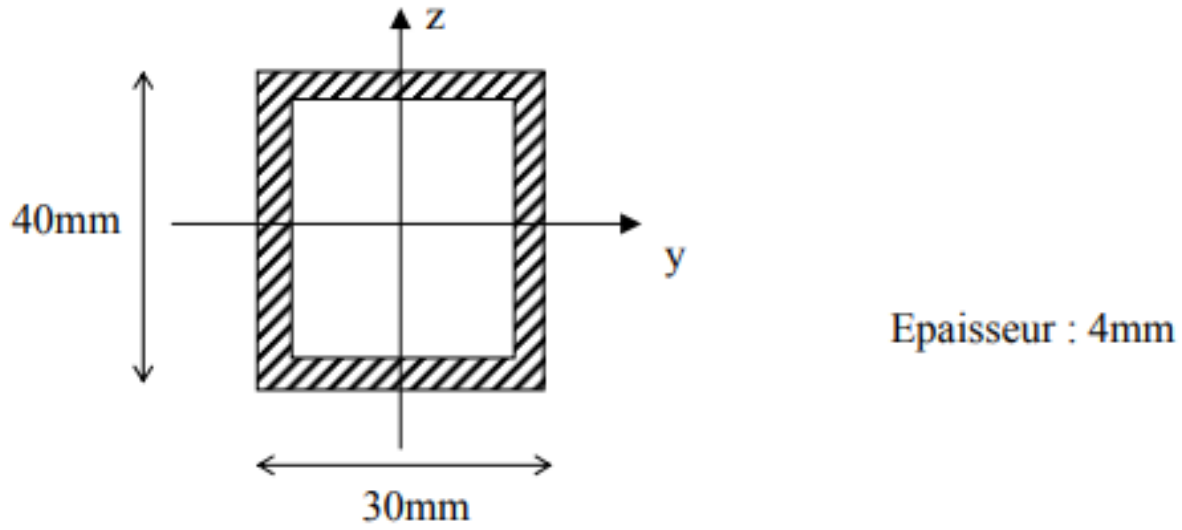


FIGURE 1 – Beam's Section Schematic

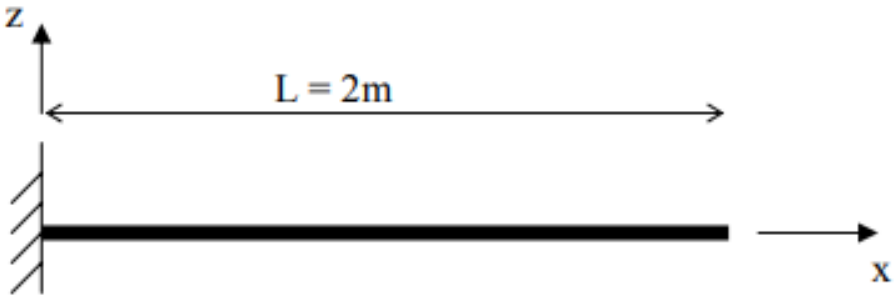


FIGURE 2 – Beam's Schematic

Since the most significant deformations are along the x-axis, we can reduce the problem to a one-dimensional one to study our case.

Here are the characteristics of the steel we need :

$$\begin{aligned} \rho &= 7,8 \times 10^{-6} \text{kg/mm}^3 \\ E &= 2.1 \times 10^5 \text{ MPa} \\ \nu &= 0.33 \\ \sigma_{R0.2} &= 660 \text{ MPa} \\ \sigma_{RM} &= 849 \text{ MPa} \end{aligned}$$

2 Tensile test

2.1 Analytical calculation :

The objective of this section is to calculate internal forces, deformations, and stresses. The normal stress equation gives us the following :

$$\sigma(x) = \frac{F}{S} \tag{1}$$

Area calculation :

$$S = 40 \times 30 - 32 \times 22 = 496\,\text{mm}^2$$

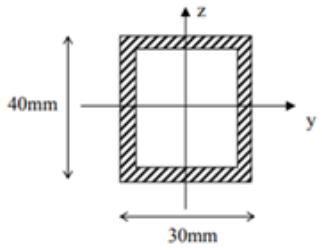


FIGURE 3 – Beam's Section

Then with de Hooke's Law :

$$\sigma(x) = E * \epsilon(x) \implies \epsilon(x) = \frac{\sigma(x)}{E}$$

$$\epsilon(x) = \frac{F}{S * E} \tag{2}$$

And with this we can have the total Elongation δL :

$$\epsilon(x) = \frac{\delta L}{L}$$

$$\delta L = L * \epsilon(x) \tag{3}$$

Displacement calculation :

$$\epsilon(x) = \frac{\partial u(x)}{\partial x} \implies u(x) = \int \epsilon(x)dx + C = \int \frac{F}{SE}dx + C = \frac{F}{SE}x + C$$

— $u(0) = 0$ because it is embedded at x = 0
— then C= 0 and :

$$u(x) = \frac{F}{SE}x \tag{4}$$

This means that the displacement is proportional to the length of the beam

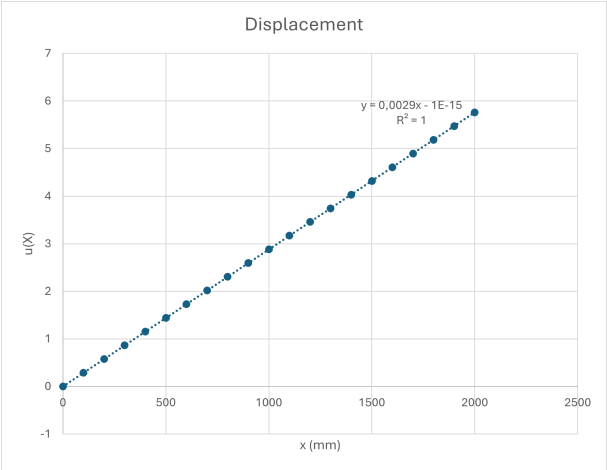


FIGURE 4 – the displacement as a function of the beam length

2.2 Analytical results

— $\sigma(x) = \dfrac{F}{S} = \dfrac{300000}{496} = 604.84\,\text{Mpa}$

— $\epsilon(x) = \dfrac{F}{SE} = \dfrac{300000}{496 \times 2.1 \times 10^5} = 2.3 * 10^{-3}$

— $\delta L = L * \epsilon(x) = 2000 * 2.3 * 10^{-3} = 4.6\,\text{mm}$

Interpretation of the results :

The applied stress is :

$$\sigma = 604.84 \text{ MPa} < \sigma_{0.2} = 660 \text{ MPa}.$$

Thus, the applied stress is lower than the conventional yield strength $\sigma_{0.2}$.

The material therefore remains in the elastic domain. No permanent plastic deformation occurs.

2.3 Nastran/Patran modeling

In this section, we will detail the construction of the model to avoid doing so in the second part.

2.3.1 Geometry modeling

To create the geometry, we need to create two different points separate by 2m and then create a curve with the 2 points created

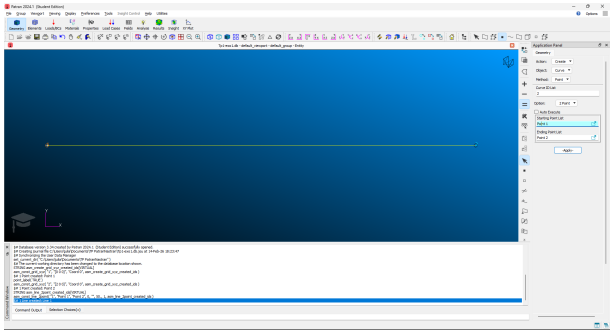


FIGURE 5 – Geometry of beam

2.3.2 Mesh modeling

In this part we initiate the mesh by requiring the software to only mesh one element and then we apply it on the curve created

2.3.3 Initial Condition

Now we applied the initial condition :

— Embedding at $x = 0$
— 300 000 N at $x = 2\,\text{m}$ along the x axis

2.3.4 Define materials

This part is for define the materials proprieties by using these values :

$$\begin{aligned} \rho &= 7,8 \times 10^{-6} \text{kg/mm}^3 \\ E &= 2.1 \times 10^5 \text{ MPa} \\ \nu &= 0.33 \end{aligned}$$

Because our problem is only along the X-axis, we are not working with a beam but a rod.

2.3.5 Results

Here is the result for the displacement, translational $(u(x))$:

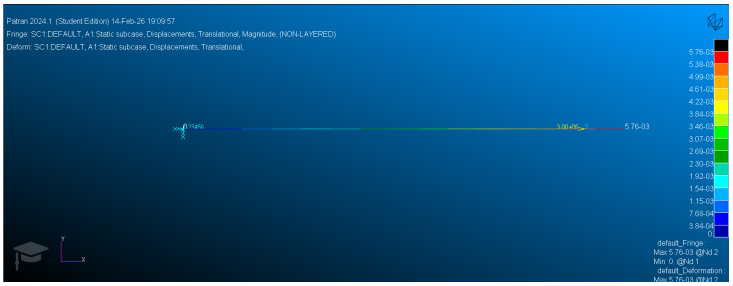


FIGURE 6 – Axial displacement field

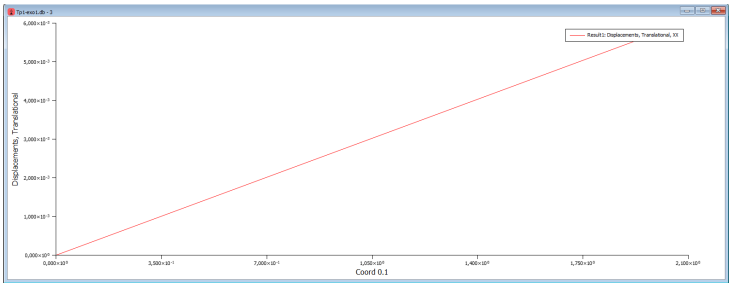


FIGURE 7 – Axial displacement as a function of beam length

We can read that the results are $u(0) = 0$ and $u(L) = \delta L = 6mm$

The result of stress :

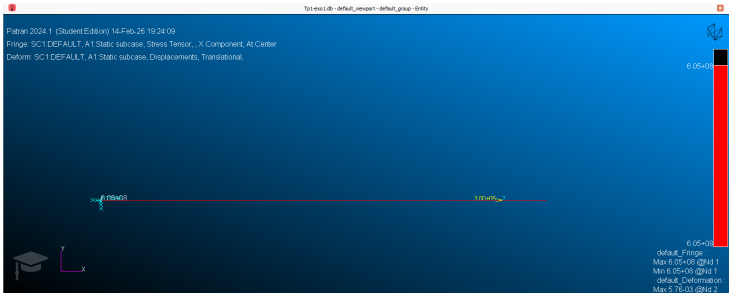


FIGURE 8 – Normal stress distribution

We can see that the stress tensor is constant and $\sigma = 605Mpa$

2.4 Interpretation of the results

Now we can compare the results from the analytical part and the result from Nastran/Patran.

2.4.1 Displacement

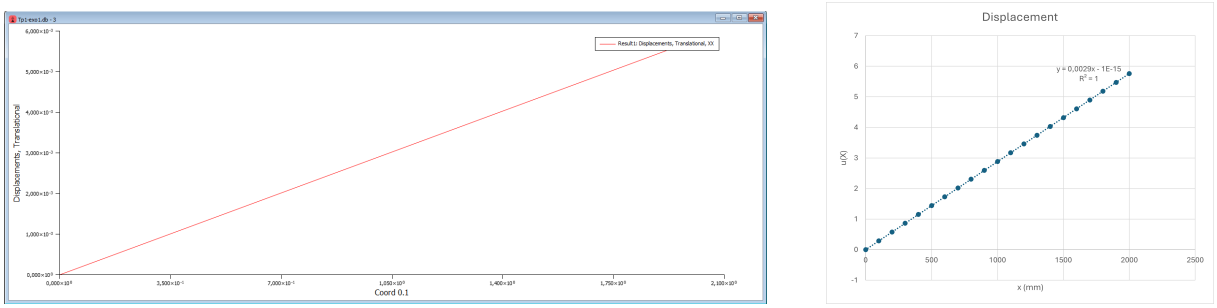


FIGURE 9 – Comparison of displacement results

On the left hand graph we have a slope of 3×10^{-3} with $u(0) = 0$ and $u(L) = 6 \times 10^{-3} mm$ On the right hand graph we have a slope of 2.9×10^{-3} with $u(0) = 0$ and $u(L) = 5.8 \times 10^{-3} mm$ we can see that the values are really close from the Nastran/Patran model and analytical model

2.4.2 Stress

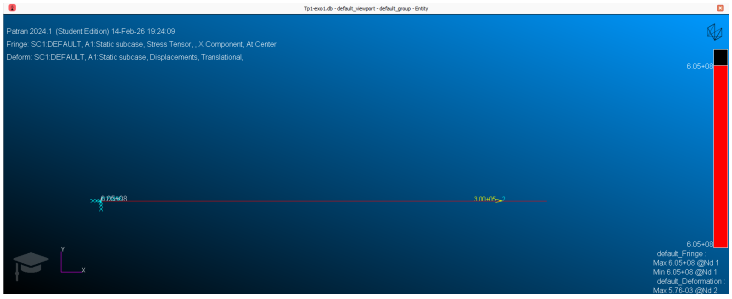


FIGURE 10 – Normal stress

We can read on the Figure 10 that the stress is at 605 Mpa and we had calculate analytically $\sigma = 604.84 Mpa$ which is really close from the model of Nastran/Patran

2.4.3 Conclusion :

The objective of this study was to compare the analytical solution of the beam problem with the numerical results obtained using Nastran/Patran.

The comparison of the displacement curves shows a very good agreement between both approaches. The slopes of the two graphs are very close , and the final displacement values differ only slightly. This small difference can be explained by numerical approximations, and mesh refinement in the finite element model.

Regarding the stress analysis, the numerical value obtained from Nastran/Patran is $605 MPa$ while the analytical calculation gives $604.84 MPa$ The difference is negligible, confirming the validity and accuracy of the numerical simulation.

In conclusion, the finite element model accurately reproduces the analytical behavior of the beam. The results validate both the theoretical formulation and the numerical implementation, demonstrating the reliability of the Nastran/Patran model for this type of structural analysis.

3 Flexion test

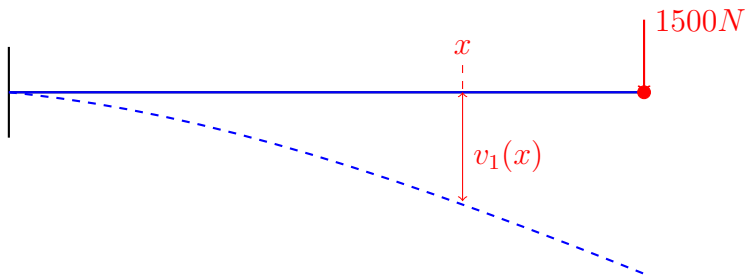
3.1 Analytical calculation

In this exercise the beam is subjected to two forces : the wheight force and a force at x = 2m along the -Y axis.

the objectif stay the same : calculate internal forces, deformations, and stresses.

It would also be interesting to divide our problem into 3 cases : first, only force ; then only weight ; and finally, force and weight at the same time. According to the superposition principle, the results of case 3 should be the sum of cases 1 and 2 let's show this.

3.1.1 Case 1 : Only the Force



The fundamental principle of statics said that :

$$\begin{cases} \sum \vec{F} = \vec{0} \implies R_0 - F = 0 \implies R_0 = F \\ \sum \vec{M_0} = \vec{0} \implies M_0 - F * L = 0 \implies M_0 = F * L \end{cases}$$

car Il n'y a aucune charge répartie le long de la poutre.

$$\frac{dV_1(x)}{dx} = -q = 0$$

Ainsi $V_1(x) = C_1$ or $V_1(L) = F$ donc :

$$V_1(x) = F$$

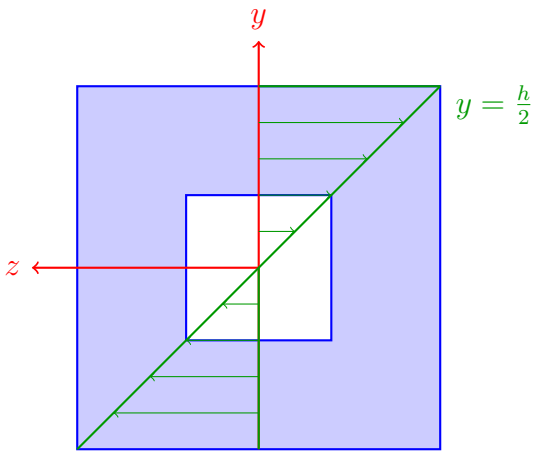
Le moment de flexion décroit en fonction de x

$$\int_x^L V(x)\,\mathrm{d}x = M_f(x)$$

$$M_{f1} = F(L - x)$$

now we have to calculate the internal forces :

$$\sigma(x, y) = \frac{-y * M_f(x)}{I}$$



The inertial moment around the axis Z is :

$$I = I_{Gcarré} - I_{Pcarré}$$

$$I = \frac{0.04*0.03^3 - 0.032*0.022^3}{12} = 6.16 \times 10^{-8} Kg.m^2$$

Then : $\sigma(x) = -4.9 \times 10^8 * (2 - x)$

$$\frac{d^2v(x)}{dx^2} = -\frac{M_f}{E*I} \implies \frac{d^2v(x)}{dx^2} = -\frac{F(L-x)}{E*I} \implies \frac{dv(x)}{dx} = -\frac{FL}{E*I}x + \frac{F}{2*E*I}x^2 + C_1$$

$$\implies v(x) = -\frac{FL}{2*E*I}x^2 + \frac{F}{6*E*I}x^3 + C_1 x + C_2$$

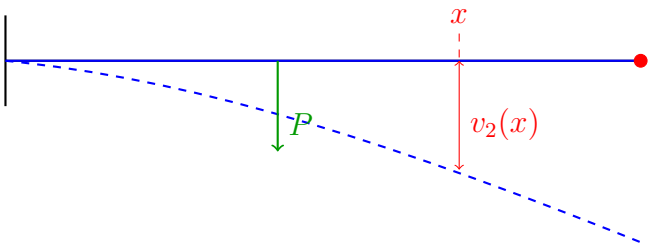
because it is embedded at x = 0 :

- $v(0) = 0$
- $v'(0) = 0$
- then $C_1 = 0$ and $C_2 = 0$:

$$v_1(x) = -\frac{F * x^2}{6 * E * I}(3L - x)$$

- $v_1(0) = 0$
- $v_1(L) = v_{max} = -309mm$:

3.1.2 Case 2 : only the weight



Now the principle is the same but with the weight :

$$\frac{dV(x)}{dx} = -q$$

or $q = \rho * S * g = 38.2N/m$ So $V(x) = -qx + C_1$

$V_2(0) = qL \implies C_1 = qL$

$$V_2(x) = q(L - x)$$

Now we have to calculate M_f

$$M_f = \int q(L-x)dx = q(Lx - \frac{x^2}{2}) + C_1$$

but $M(L) = 0 \implies C_1 = -\frac{qL^2}{2}$

so :

$$M_{f2} = q(Lx - \frac{x^2}{2} - \frac{L^2}{2})$$

then

$$\frac{d^2v(x)}{dx^2} = -\frac{M_f}{E*I} \implies \frac{d^2v(x)}{dx^2} = -\frac{q(Lx-\frac{x^2}{2}-\frac{L^2}{2})}{E*I} \implies \frac{dv(x)}{dx} = \frac{-q}{EI}(\frac{Lx^2}{2} - \frac{x^3}{6} - \frac{L^2x}{2} + C_1)$$
$$\implies v(x) = \frac{-q}{EI}(\frac{Lx^3}{6} - \frac{x^4}{24} - \frac{L^2x^2}{4} + C_1x + C_2)$$

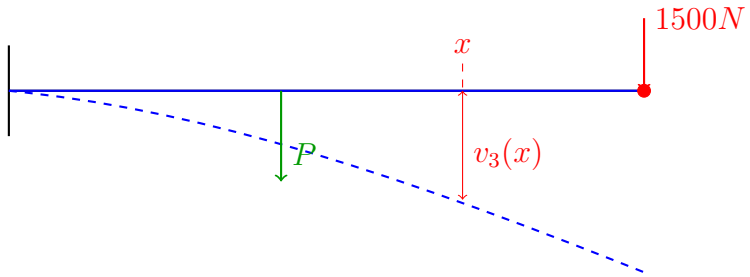
because it is embedded at x = 0 :

- $v(0) = 0$
- $v'(0) = 0$
- then $C_1 = 0$ and $C_2 = 0$:

$$v_2(x) = \frac{-qx^2}{EI}(\frac{Lx}{6} - \frac{x^2}{24} - 1)$$

- $v(0) = 0$
- $v(L) = v_{max} = 6mm$

3.1.3 Case 3 : Force + Weight



Now the principle is the same but with the force and the weight :

$$\frac{dV(x)}{dx} = -q$$

or $q = \rho * S * g = 38.2N/m$ So $V(x) = -qx + C_1$

$V_3(L) = F \implies C_1 = F + qL$

$$V_3(x) = F + q(L-x)$$

then :

$$M_f = Fx + q(Lx - \frac{x^2}{2}) + C_1$$

but $M(L) = 0 \implies C_1 = -FL - \frac{qL^2}{2}$

$$M_f = Fx + q(Lx - \frac{x^2}{2}) - FL - \frac{qL^2}{2}$$

$$M_{f3} = F(x - L) + q(Lx - \frac{x^2}{2} - \frac{qL^2}{2}) = M_{f1} + M_{f2}$$

To calculate the dispalcement we use that the integral is linear then we have $v_3(x) = v_1(x) + v_2(x)$

$$v_3(x) = -\frac{F * x^2}{6 * E * I}(3L - x) - \frac{qx^2}{EI}(\frac{Lx}{6} - \frac{x^2}{24} - 1)$$

— $v(0) = 0$
— $v(L) = v_{max} = -315mm$

3.2 Analytical results

So the objectif of this part is to verify the supeposition principle.

3.2.1 Force

Displacement :

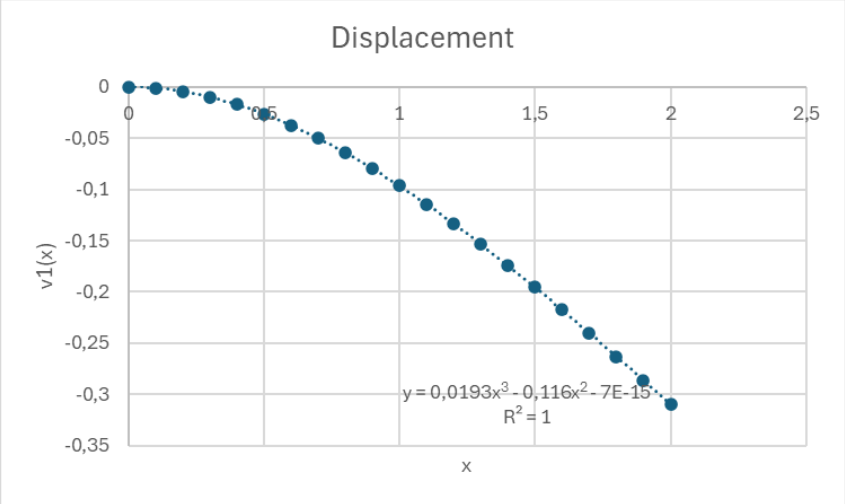


FIGURE 11 – Displacement (Force)

The curve shows a progressive increase of displacement along the beam, reaching its maximum at the free end, as expected for a cantilever under bending.

Stress :

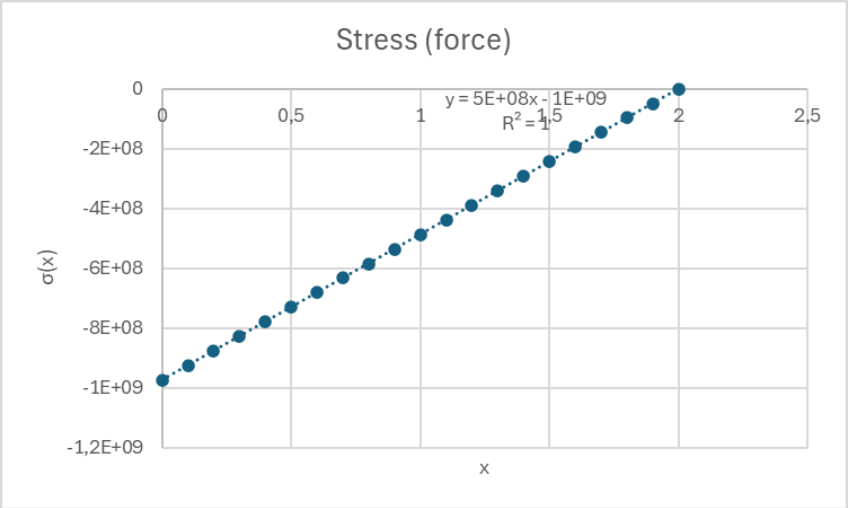


FIGURE 12 – Bending stress (Force)

The stress also increases toward the fixed support, where the bending moment is maximum

Bending and reaction :

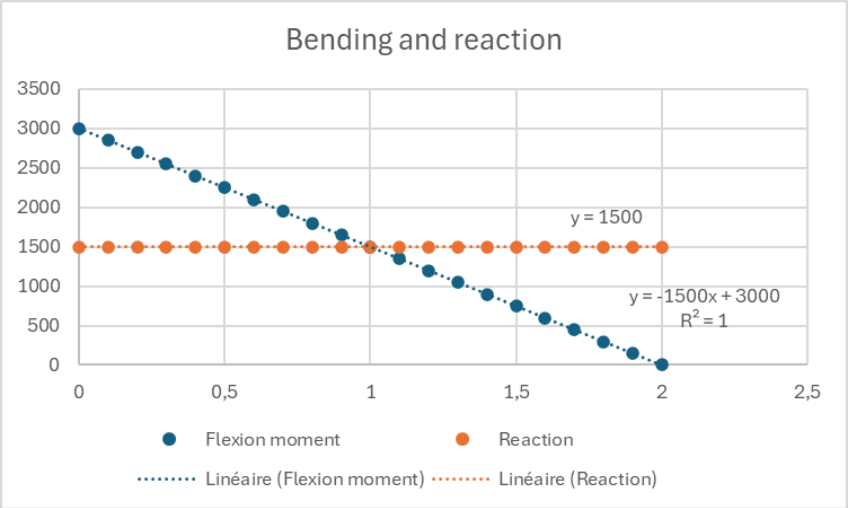


FIGURE 13 – Bending and reaction (Force)

The bending moment decreases linearly along the beam, while the reaction at the support remains constant.

3.2.2 Weight

Displacement :

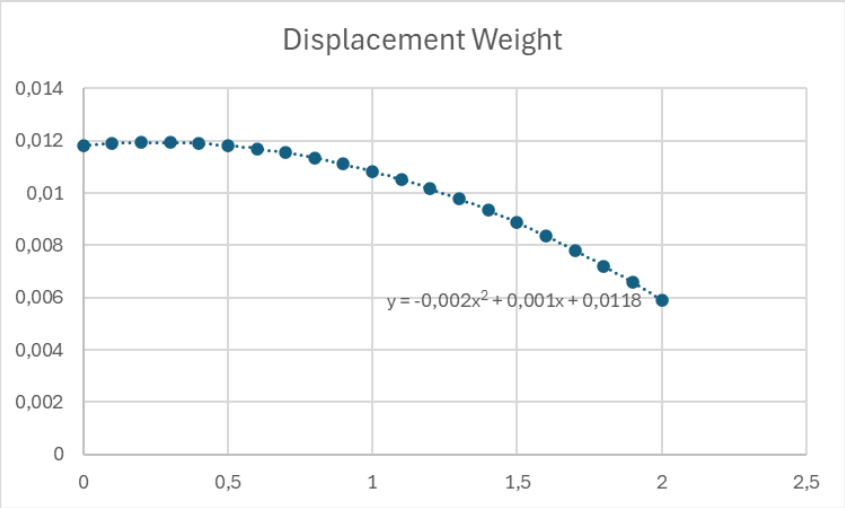


FIGURE 14 – Displacement (Weight)

The displacement follows a smooth parabolic shape with a maximum at the free end.

Stress :

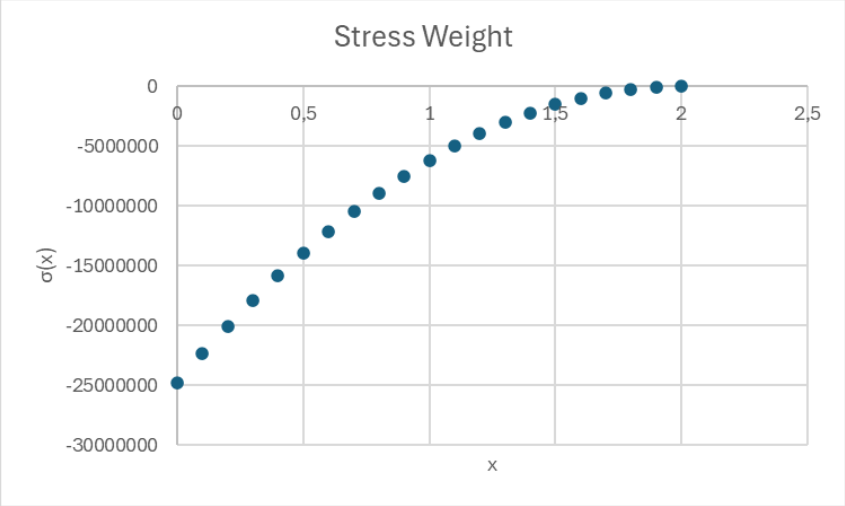


FIGURE 15 – Stress (Weight)

The stress increases toward the fixed support due to the effect of the distributed load.

Bending and reaction :

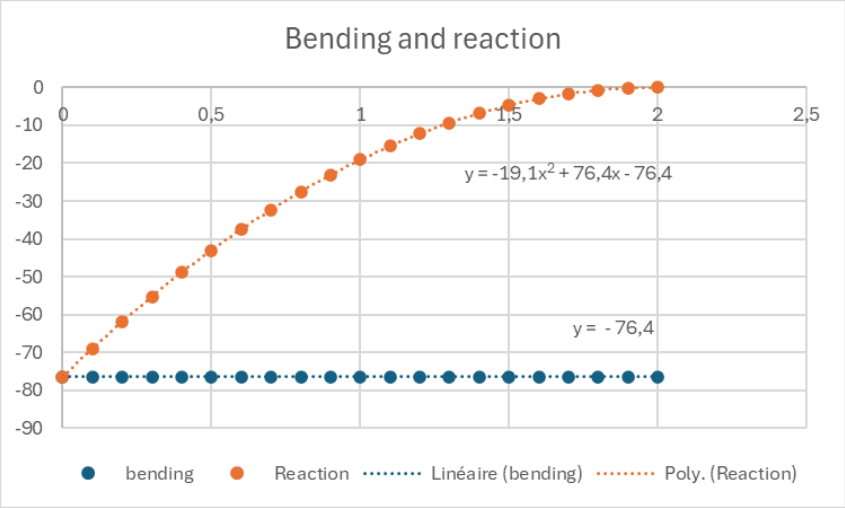


FIGURE 16 – Bending and reaction (Weight)

The bending moment decreases along the beam, while the reaction at the support remains constant, as predicted by equilibrium.

3.2.3 Force + Weight

Displacement :

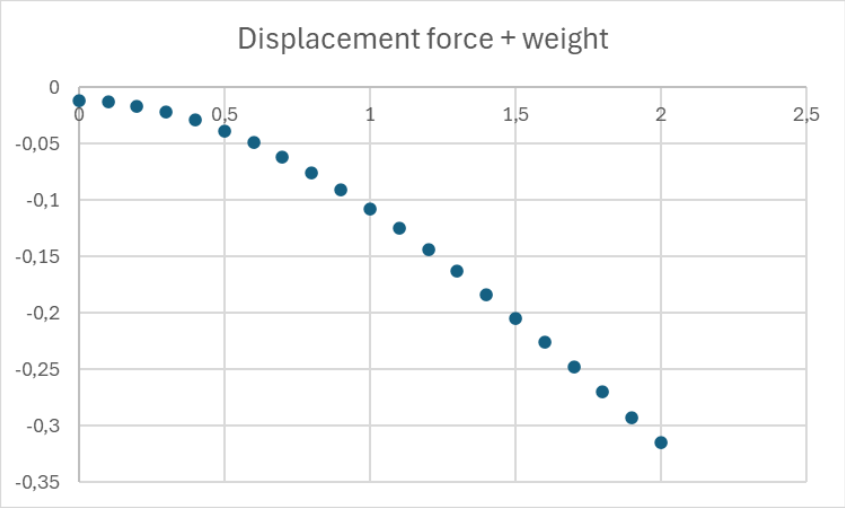


FIGURE 17 – Displacement (Force + Weight)

Stress :

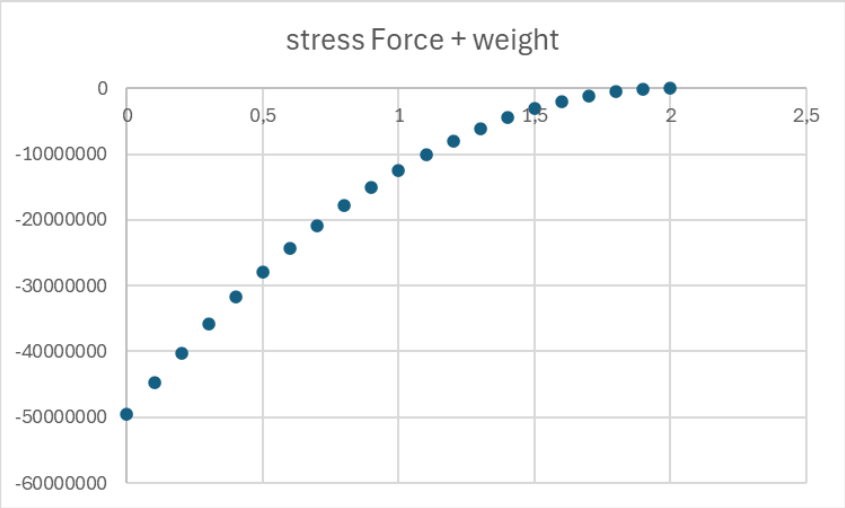


FIGURE 18 – Stress (Force + Weight)

Bending and reaction :

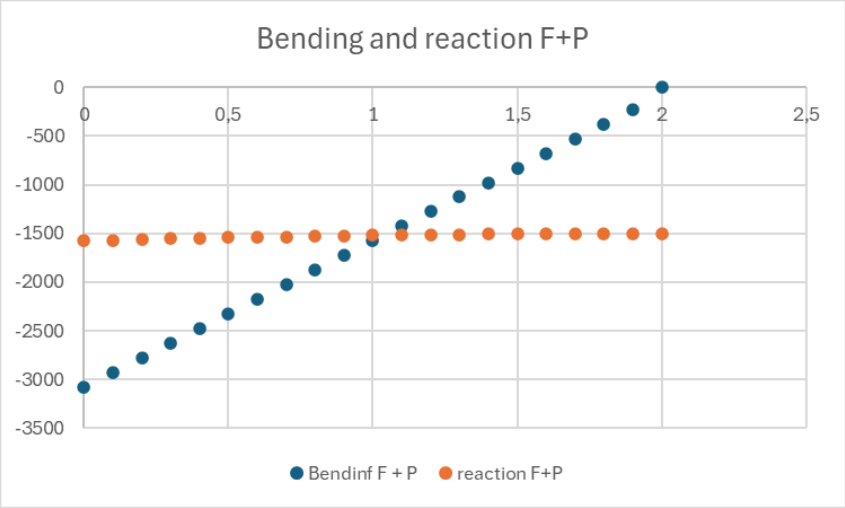


FIGURE 19 – Bending and reaction (Force + Weight)

The bending moment and reaction values correspond to the sum of the two individual cases, confirming linear behavior.

3.3 Nastran/Patran modeling

3.3.1 Geometry modeling

Nothing different here compared to the part 2.3.1

3.3.2 Mesh modeling

We have seen analytically that we have a parabolic displacement, so it is wise to have a large number of elements during mesh generation ; in order to limit computation time, we will reduce this to 20 elements.

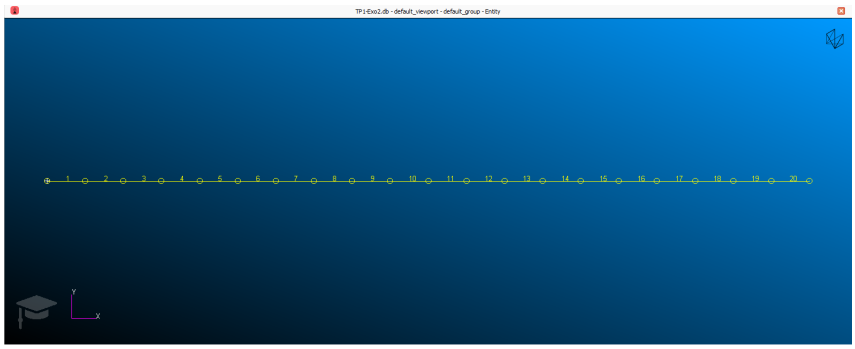


FIGURE 20 – Mesh of the Beam

3.3.3 Initial Condition

Now we applied the initial condition :

— Embedding at $x = 0$
— 1500 N at $x = 2\,\text{m}$ along the -Y axis

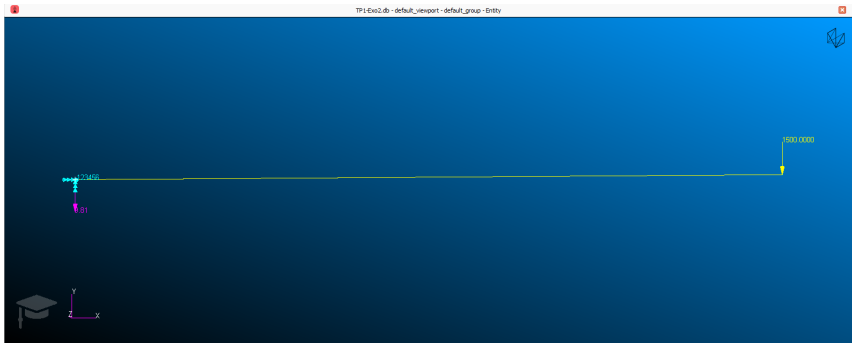


FIGURE 21 – Initial conditions

3.3.4 Define materials

Nothing different here compared to the part 2.3.4 except that now we have a beam because we are working in the (X,Y) plane.

Here is the information from its section :

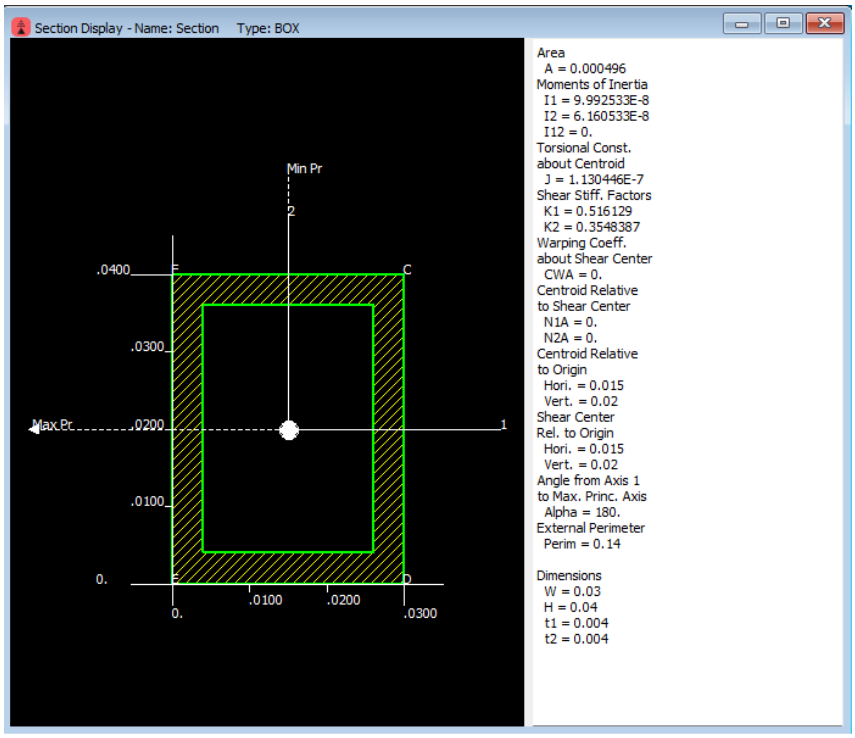


FIGURE 22 – Beam's Section information

3.3.5 Results

Case 1 : Only the force :

We can see on the following figure that the maximum displacement is 309mm

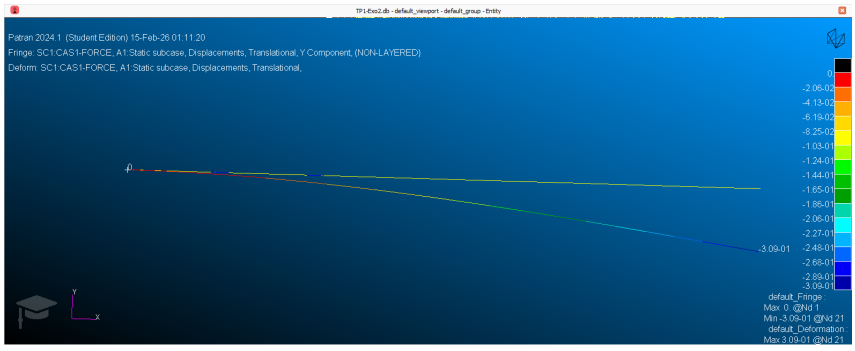


FIGURE 23 – Reaction force

The reaction to embedding is 1500N.

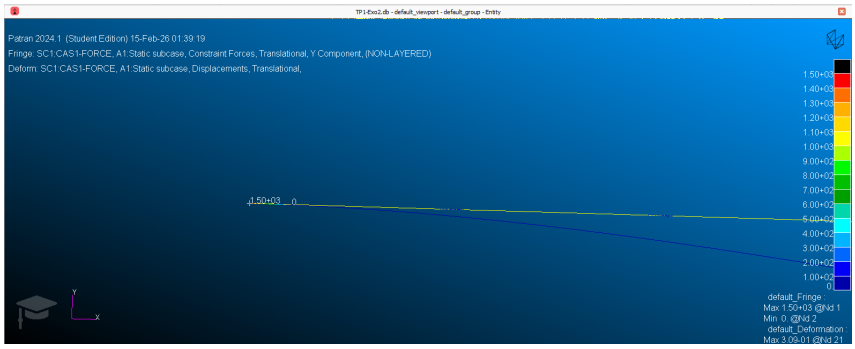


FIGURE 24 – Bending moment

The torque at the joint is 3000 Nm.

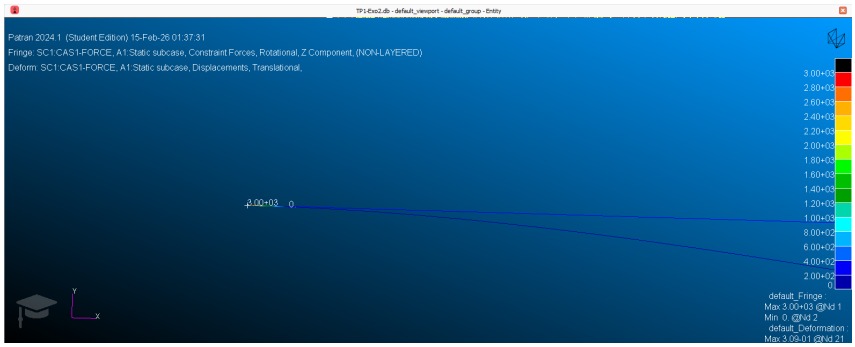


FIGURE 25 – Transverse displacement

The bending stress is given by the following graph and depends on x :

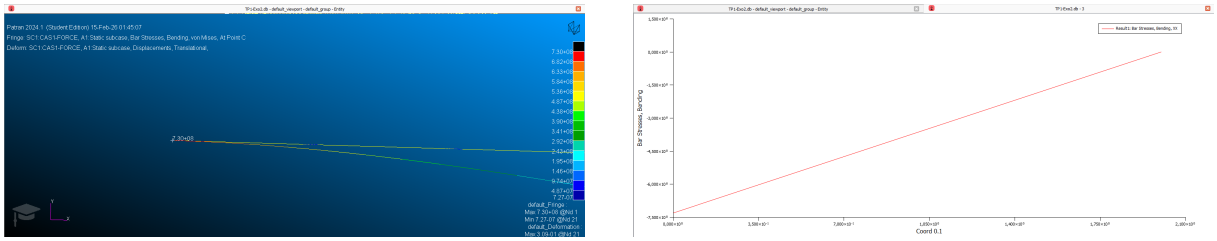


FIGURE 26 – Stress

Case 2 : Only the weight force :

We can see on the following figure that the maximum displacement is 6mm

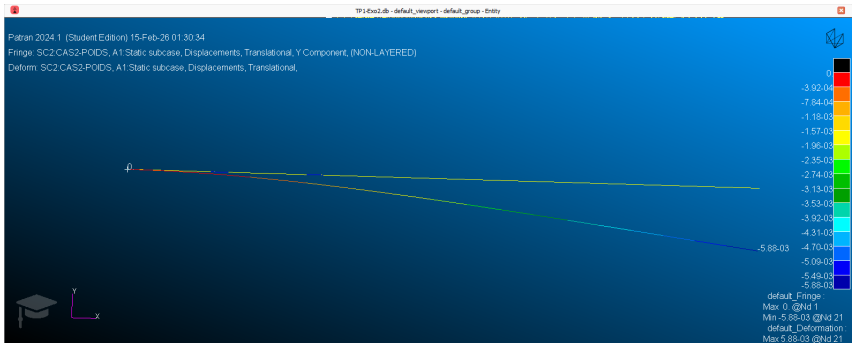


FIGURE 27 – Bending stress distribution

The reaction and moment at the embedding point is 76N.

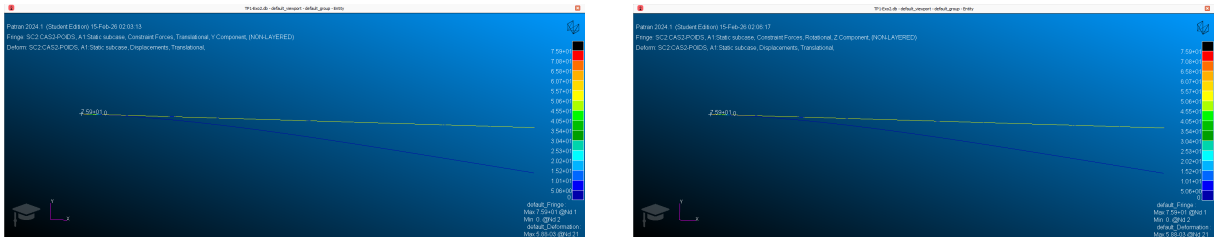


FIGURE 28 – Stress

The bending stress is given by the following graph and depends on x :

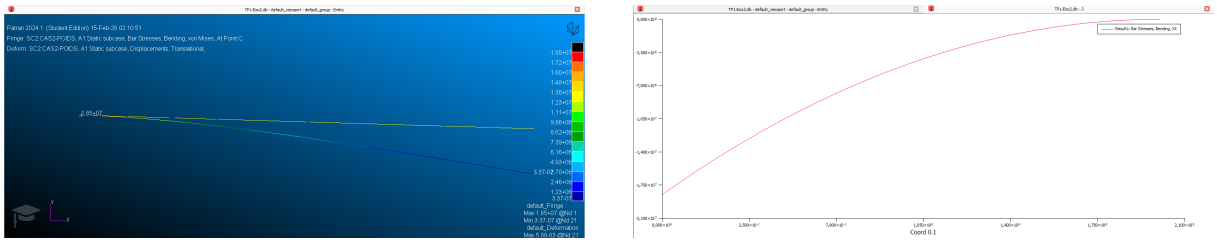


FIGURE 29 – Stress

case 3 : Force + weight force

We can see on the following figure that the maximum displacement is 315mm

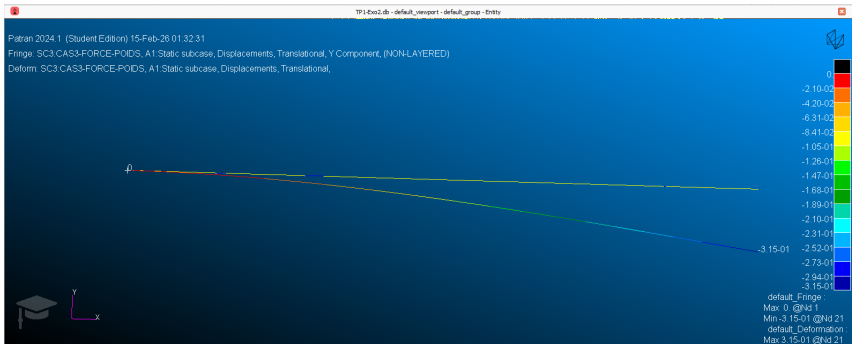


FIGURE 30 – Reaction force

The reaction force at the mounting point is 1580N.

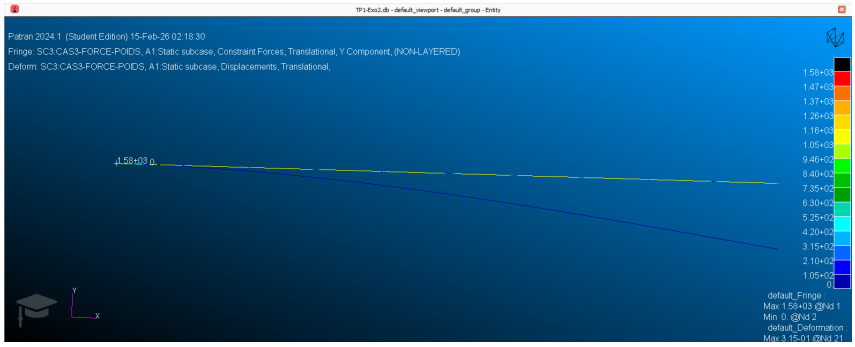


FIGURE 31 – Bending moment

The moment at the embedding point is 3080N.

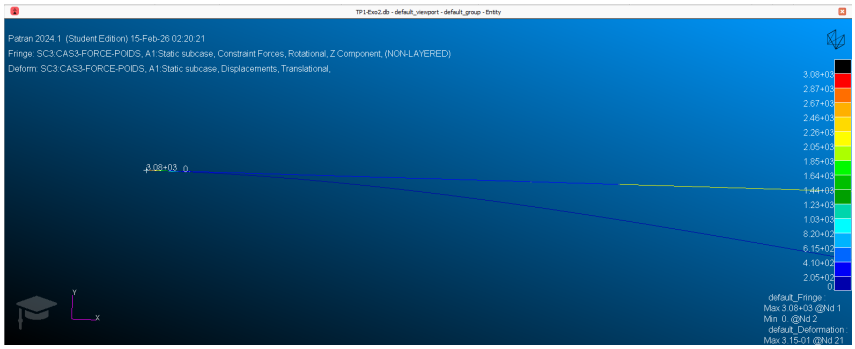


FIGURE 32 – Bending stress distribution

The bending stress is given by the following graph and depends on x :

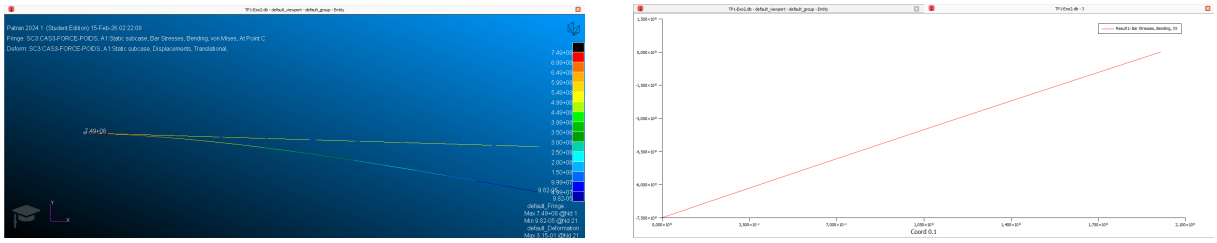


FIGURE 33 – Stress

3.4 Interpretation of the results

For the problem with only applied force, the maximum displacement calculated numerically is 309 mm, the same as the analytical solution. The reaction force and the bending moment at the fixed end are also in agreement with the analytical solution of the equilibrium equation.

For the case with only the weight, the maximum displacement is 6 mm, which is again in agreement with the analytical solution. The reactions and the variation of the stress are consistent with the action of a uniformly distributed load.

For the problem where both force and weight are applied simultaneously, the maximum displacement is 315 mm, which is the sum of the two previous problems. The same is true for the reactions and the internal forces.

All of this shows that the structure is linear in the elastic domain and that the superposition principle holds.

4 General conclusion

The objective of this laboratory work was to study the linear mechanical behavior of a hollow rectangular beam subjected to axial loading and bending, using both analytical methods and numerical simulations. The main purpose was to compare theoretical solutions derived from strength of materials with finite element results, and to validate the assumptions of linear elasticity and the superposition principle.

In the first part of the study, the tensile test allowed us to analyze a uniaxial loading case. The analytical calculations provided expressions for stress, strain, and displacement along the beam, showing a uniform normal stress distribution and a linear displacement profile. The numerical model confirmed these results with very good accuracy. The small differences observed between analytical and numerical displacements can be attributed to numerical approximations and discretization effects inherent to the finite element method. Moreover, the maximum stress remained below the yield strength of the material, confirming that the beam operates entirely within the elastic domain.

The second part of the work focused on the flexion test, where the beam was subjected to transverse loading due to an external force, its own weight, and a combination of both. Analytical solutions were derived for each case using equilibrium equations and beam

theory. The resulting displacement fields exhibited the expected parabolic shape, and the stress distributions followed classical bending behavior. The numerical simulations reproduced these results accurately, both in terms of maximum displacement and internal forces.

A key outcome of this study is the verification of the superposition principle. The results obtained for the combined loading case were equal to the sum of the individual cases (force only and weight only), both analytically and numerically. This confirms the linearity of the system and validates the use of linear elastic theory for this type of structural problem.

Overall, the close agreement between analytical calculations and numerical simulations demonstrates the reliability and accuracy of the finite element modeling approach when properly implemented. This work highlights the importance of analytical solutions as a reference for validating numerical models, while also showing the efficiency of numerical methods for solving more complex structural problems. The methodology used in this TP can therefore be confidently extended to more advanced analyses in structural mechanics.